CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

Nov/Dec 2013

EE206 – Digital Electronics & microprocessor

Date: 15/11/2013, Friday | **Time: 01:30 p.m. To 04:30 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) State and explain De Morgan's theorems with truth table and symbol. [05]

(b) State and explain the distributive law. [02]

Q-2 (a) Explain the working of a full adder circuit? Draw its block diagram, logic diagram and truth table. [06]

(b) Implement the logic function of the below expression using 8:1 multiplexer. [06]
$F(A,B,C,D) = \sum m(1,3,4,6,7,10,14)$

(c) What is the difference between synchronous and asynchronous counter? [02]

OR

Q - 2 (a) Give the comparison of CMOS and TTL [06]

(b) With neat block diagram explain the function of an encoder. [06]

(c) What is the function of a shift register ? Give its applications. [02]

Q - 3 (a) Explain with diagram the working of Positive Edge Triggered D flip-flop . Give its truth table. [06]

(b) Explain with truth table a BCD to 7-segment decoder. [06]

(c) Draw the structure of three input K-map. [02]

OR

Q-3 (a) Fill in the blanks [06]

i) The ________ and ________ gates are called as " universal Gates"
ii) $A + A' =$ ________
iii) Output of EX-NOR with inputs A and B is ________.
iv) A multiplexer is also known as ________.
v) The ________ circuits do not use any memory.
vi) Obtain the 2's complement of $(1011\ 1000)_2$

(b) Prove the following identities: [06]

(i) $(A+B'+AB)(A+B)A'B' = 0$
(ii) $A+A'B+AB = A+B$
(iii) $AB' + A'B + AB + A'B' = 1$

(C) Using a Karnaugh map, reduce the expression $Y = A + AB'C + AB$. [02]

SECTION – II

Q-4 (a) Draw the diagram of 8085 Bus Structure and explain address bus, data bus and control bus in detail. [05]

(b) Explain the difference between a machine language and assembly language. [02]

Q-5(a) Explain addressing modes of 8085 microprocessor with suitable examples. [07]

(b) Explain 8085 De-multiplexed Address and Data Bus with control signals with necessary diagram [07]

OR

Q-5(a) What is flag? Draw the format of 8085 flag register and explain in detail. [07]

(b) Explain the function of following instructions. [07]

(i) MVI M, data (ii) ADD M (iii) XCHG
(iv) CMA (v) DAD rp (vi) DCR M (vii) ORA M.

Q-6 (a) Write 8085 microprocessor based assembly program for any three of the following. Explain the program logic in brief. [12]

(i) Write a program for Exchange the contents of memory location 2000H and 4000 H.

(ii) Write a program Find the Two's complement of a 16-bit number.

(iii) Write a program to Shift an 8-bit number left by 2bits.

(iv) Write a program to find the smallest number in a data array. The numbers of a series are : 86 , 58 and 75.

(b) Define opcode and operand and specify the opcode and operand in the instruction MVI H , 47H [02]

OR

Q-6 (a) Write a program to count from 0 to 9 with a one-second delay between each count. At the count of 9, the counter should reset itself to 0 and repeat the sequence continuously. Use register pair HL to set up the delay, and display each count at one of the output ports. Assume clock frequency of the microcomputers is 1 MHz. [07]

(b) Give the classification of memory and explain in brief [07]